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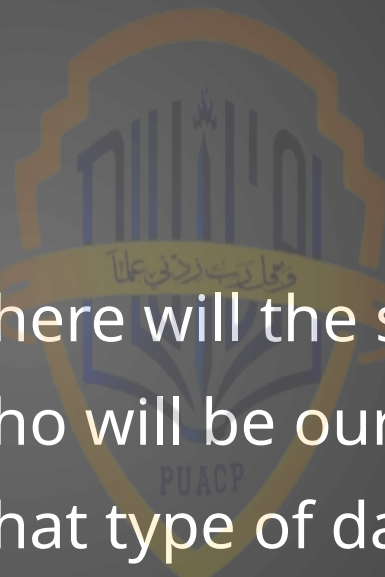
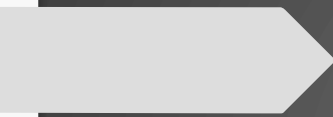
Research Design-I

Lecture by: Dr. Iqra Abdullah

Book Reference: Chapter 3: Kothari, C.R, (2008). Research methodology: Methods and techniques New Delhi, New Age International Publishers

Research Design

- ▢ A research design is the arrangement of conditions for collection and analysis of data in a manner that aims to combine relevance to the research purpose with economy in procedure (Kothari, 2008).
- ▢ It is a plan that specifies the sources and types of information relevant to the research problem.
- ▢ It is a strategy specifying which approach will be used for gathering and analysing the data.
- ▢ It also includes the time and cost budgets since most studies are done under these two constraints.



Research Design

1. Where will the study be carried out?
2. Who will be our target population?
3. What type of data is required?
4. Where can the required data be found?
5. What periods of time will the study include?
6. What will be the sample design?
7. What techniques of data collection will be used?
8. How will the data be analysed?
9. In what style will the report be prepared?

Parts of Research Design



Sampling
Design



Observational
Design



Operational
Design

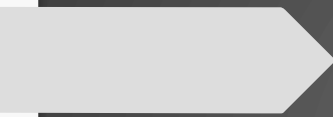


Statistical
Design

Why should we develop a research design?

- ▢ Smooth Sailing
- ▢ Maximal information → minimal effort, time and money
- ▢ Constitutes foundation
- ▢ Organization of ideas
- ▢ Identification of flaws and inadequacies
- ▢ Critical Analysis
- ▢ Reliability of the results
- ▢ Avoid misleading conclusions





Characteristics of good research design

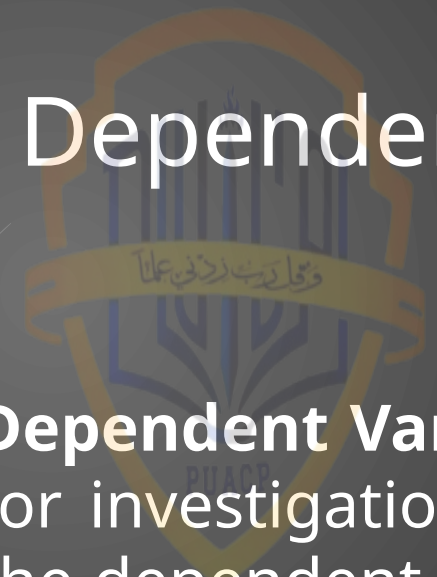
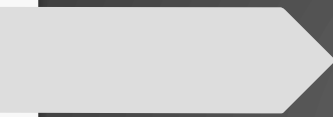
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- ▣ Flexible
 - ▣ Appropriate
 - ▣ Accurate
 - ▣ Efficient
 - ▣ Economical
 - ▣ Minimal Bias
 - ▣ Minimal Error
 - ▣ Maximises the reliability of the data
 - ▣ Maximal information



Concepts Research Design

Variables

- ▢ **Variable:** Anything that varies or changes from one instance to another; can exhibit differences in value, usually in magnitude or strength, or in direction.
 - ▢ The dependent variable (also known as the criterion variable).
 - ▢ The independent variable (also known as the predictor variable).
- ▢ **Constant:** Something that does not change; is not useful in addressing research questions.



Dependent Variables and Independent Variables

Dependent Variable: It is the main variable that lends itself for investigation as a viable factor. Through the analysis of the dependent variable (i.e., finding what variables influence it), it is possible to find answers or solutions to the problem.

Independent Variable: It is the one that influences the dependent variable in either a positive or negative way.

Examples

If you will get education, you will become a good citizen of Pakistan

Education



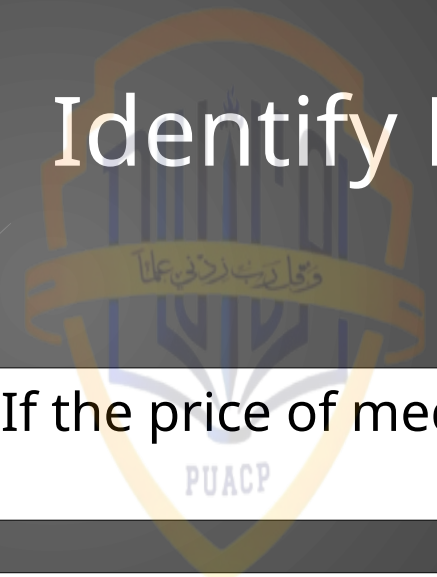
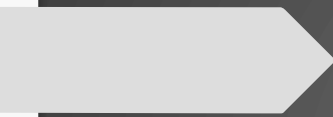
Good Citizen

If you will get education, your income level will increase

Education



Income



Identify Dependent and Independent Variables

If the price of medicines will increase, it will lead to decrease in the demand of medicines.

When interest rates are high, people tend to make less investment.

When income grows, consumers purchase more goods and services.

Extraneous Variable

- ▢ Independent variables that are not related to the purpose of the study, but may affect the dependent variable are termed as extraneous variables.
 - ▢ Example: There is a relationship between achievement in social studies and their confidence. In this case confidence is an independent variable and social studies achievement is a dependent variable.
 - ▢ Intelligence may as well affect the social studies achievement, but since it is not related to the purpose of the study undertaken by the researcher, it will be termed as an extraneous variable.
- ▢ **Control:** The technical term 'control' is used when we design the study minimising the effects of extraneous independent variables
- ▢ **Confounded Relationship:** When the dependent variable is not free from the influence of extraneous variable(s), the relationship between the dependent and independent variables is said to be confounded by an extraneous variable(s).



Identify Dependent and Independent Variables of your Project



Hypothesis Formulation

- When a prediction or a hypothesised relationship is to be tested by scientific methods, it is termed as research hypothesis. The research hypothesis is a predictive statement that relates an independent variable to a dependent variable.

Eating breakfast every morning improves concentration throughout the day.

If I water my plants regularly, then they will grow taller

Drinking a cup of coffee before bed leads to difficulty falling asleep.

Other Research Terminologies

- ▮ **Experimental/Test and Control Groups:** When a group is exposed to usual conditions, it is termed a 'control group', but when the group is exposed to some special condition, it is termed an 'experimental or test group'.
- ▮ **Treatments:** The different conditions under which experimental and control groups are put are usually referred to as 'treatments'

Other Research Terminologies

- ▮ **Experiment:** The process of examining the truth of a statistical hypothesis, relating to some research problem, is known as an experiment.
 - ▮ Example, we can conduct an experiment to examine the usefulness of a certain newly developed drug. Experiments can be of two types viz., absolute experiment and comparative experiment.
- ▮ **Experimental/Test Unit:** The pre-determined plots or the blocks, where different treatments are used, are known as experimental units or test units.



Hypothesis Testing

